

A Term Paper
Report On
A Multimodal Deep Learning System for Skin Cancer Detection
Using Generative AI

Submitted to



Amity University, Uttar Pradesh

In partial fulfillment of the requirements for the award of the degree
of Bachelor of Technology

(IT)

By

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DECLARATION

I, Parteek Vashisht, student of B.Tech (IT) hereby declare that the project titled “A Multimodal Deep Learning System for Skin Cancer Detection Using Generative ” which is submitted by me to *Department of information technology,ASET* , Amity University Uttar Pradesh, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology(IT) , has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

Noida

Date

Name and Signature of Student

CERTIFICATE

On the basis of declaration submitted by Parteek Vashisht, student of B. Tech (*Discipline*), I hereby certify that the project titled “A Multimodal Deep Learning System for Skin Cancer Detection Using Generative AI” which is submitted to *Department of Information Technology, ASET*, Domain of Engineering and Technology, Amity University Uttar Pradesh, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology (*IT*), is an original contribution with existing knowledge and faithful record of work carried out by him/them under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

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Parteek Vashisht
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Abstract

Cancer on the skin and most importantly a skin cancer called melanoma is one of the most prevalent cancers in the world and is very dangerous especially when not detected at an early age. Traditional diagnosis techniques. Despite their solidity, the traditional approaches to diagnosis are most vulnerable with the ratio in the collection of data and subjectivity and inclusion of many groups drastically inadequate. The emergence of artificial intelligence and, in particular, the deep learning and generative AI technologies have introduced the new cluster of directions related to the enhanced diagnostic algorithms in skin cancer. In this studious survey, the scientists appraise Generative Artificial Intelligence (GenAI) that is combined with multimodal deep learning systems to address the incumbent diagnostic complexity. The presented article systematically reviews the proposed methods, i.e., Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models, regarding the task of collecting dermoscopic image synthesis and multimodal data fusion. Their selected research questions are those that touch on improvement of datasets, best fusion architectures, improvement of accuracy of diagnosis, alignment in data as well as explainability in clinical settings. Despite the positive emerging progress, several challenges exist including access to high quality datasets, synthesized data validation, and regulatory mechanisms. In short, it can be mentioned that this review concludes with the projected future research study and clinical practices that can present the differences between technological development and dermatological feasibilities.

Introduction

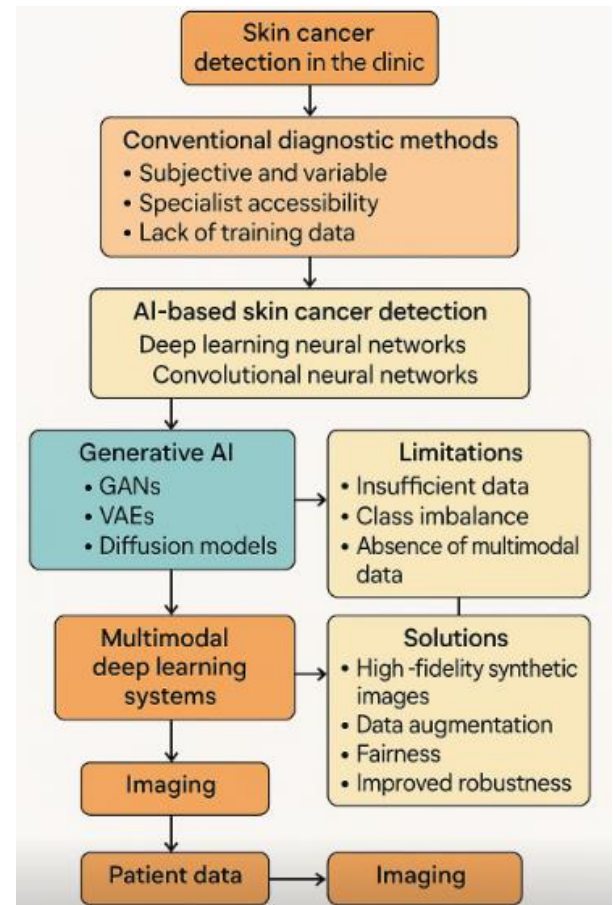
The Skin cancer is one of the most common and deadliest cancers that occurs in the entire world and melanoma and non-melanoma type of skin cancer comprise a high percentage of dermatology morbidity and mortality rate in case it is not diagnosed at the early stage [1]. The recent traditional diagnostic pathways have drawbacks in that they are highly dependent on the experience of the dermatologists through the eye examination and dermoscopy which are plagued by inter-observer variation and resource constraints based on the availability of talented and skilled professionals especially in low-income settings. Such problems as delays in diagnosis and misclassification can cause poor patient outcomes, which means that automated and scalable solutions to diagnostic purposes are a top priority [2]. Artificial intelligence (AI) has already become a groundbreaking method in medical imaging, and convolutional neural networks (CNNs) proved to achieve dermatologist-level performance in the classification of skin cancer on curated data [3]. Nevertheless, present AI systems are generally trained using single-modal dermoscopic images, which can not necessarily represent expanded clinical contexts, including history of lesion, patient demographics, or clinical photographs, making them many more vulnerable and unstable. Also, dermatology AI models are challenged by the smallness and asymmetry of datasets, particular rare lesion types and minority skin tones, and hence show biases and diminished generalization in various clinical settings [4].

Generative AI and more specifically generative adversarial networks (GANs) and diffusion models have demonstrated particular promise in overcoming these constraints through the generation of realistic skin lesion images, the augmentation of training sets, and the greater representation of brain scans. The generation of synthetic images with GANs has already been used in skin cancer detection to overcome class imbalance and data enrichment to improve the classification accuracy and sensitivity, and to exclude the requirement of expensive manual annotations [5]. Simultaneously, multimodal deep learning solutions, which combine dermoscopic images, clinical photographs, and patient metadata, including age, sex, location and type of lesion, and skin type, give a broader diagnosis context and raise the level of the model that imitates the human holistic judgement of the dermatologist. Research findings have demonstrated that multimodal models have an advantage over unimodal models since they utilize complementary data among modalities to make better, explainable predictions when detecting skin cancer [6].

This article of research examines the utilization of various models of AI to identify the presence of lung cancer. It concentrates on models and, in addition, paper also discusses the operations of these models and speaks what the problem is to implement these models in real hospitals. The future of AI in area of health care will also be discussed in this paper and the paper will give a review of related work, then the methodology, results and conclusion will be provided.

Literature Review

The traditional sequence of the first skin cancer diagnosis included visual inspection, examination with a dermoscope, and the histopathological one. Nevertheless, this diagnostic algorithm is associated with a number of limitations contributing to artificial intelligence (AI) integration. Though the basis of visual inspection is indeed essential, it is subjective and depends on the experience of the clinician. An extensive body of work demonstrated



that there are significant differences in the accuracy of diagnosis among practitioners, especially when it comes to aggressive cancerous diseases, such as melanoma [1, 3].

Dermoscopy came as an improvement to the naked-eye inspection where magnification was used to enhance the visibility of lesions on the skin. Although it is better than visual inspection alone, dermoscopy remains strongly influenced by training and experience of the involved clinician. Experienced dermatologists have an accuracy of 75-84% as regards to the diagnosis, which indicates the necessity of searching improvements [2, 3].

Histopathological dissection, although regarded as the best because of its accuracy is invasive and gives painful sensation, some scarring as well as time consuming procedures. Such constraints are especially troublesome in low-resource environments, where a failure to diagnose in a timely manner may adversely affect patient outcomes [1].

The introduction of AI in dermatology is an all-positive paradigm shift that has gained momentum since the eighties. The older AI models were dependent on the machine learning frameworks in which they extracted predefined specifications in accordance with the dermatologic heuristic such as the ABCD rule (Asymmetry, Border irregularity, Color variation, and Diameter) [2]. Support vector machines, back-propagation neural networks, genetic algorithms, etc. had been used, but their application was limited by the labor intensive feature engineering problem which involved identifying useful image features manually [4, 5].

With the introduction of deep learning, especially Convolutional Neural Networks (CNNs), this area was disrupted. The CNNs have the hierarchical processing ability that enables them to directly learn intricate visual modes strictly based on raw picture data. This proved completely unnecessary and impractical to manually specify feature extraction and instead allowed the direct performance of CNNs to either match and in some cases surpass a performance of the dermatologists visiting a board [3, 6, 10]. Specifically, though, CNNs of Tschandl et al. were shown to achieve an expert-rate performance in their work on skin cancer classification [3, 18].

Systematic reviews and meta-analyses have directly compared the diagnostic accuracy of AI systems and human clinicians. A large meta-analysis of 19 studies found that AI had a mean sensitivity and specificity of 87.0% and 77.1%, respectively, compared to 79.8% and 73.6% for clinicians—statistically significant differences [3]. The performance difference is more pronounced when comparing to general practitioners: AI systems exhibited 92.5% sensitivity and 66.5% specificity compared to 64.6% and 72.8%, respectively, for generalists [3]. AI also performed better compared to non-specialist dermatologists in most instances, whereas performance compared to expert dermatologists (AI: 86.3% sensitivity, 78.4% specificity vs. experts: 84.2%, 74.4%) was almost identical [3].

This implies that AI is not intended to replace specialist dermatologists but to complement diagnostic abilities, particularly in less experienced hands. AI has the potential to serve as a decision-support system, enhancing diagnostic confidence in complicated or unusual cases. Consumer-directed AI products have the potential to enable patients to identify suspicious lesions at an earlier stage, leading to timely medical referral [3, 6, 10].

AI is also promising in dermatopathology by aiding in lesion classification and decreasing diagnostic variability [16]. The clinical application of AI has reached milestones with regulatory clearances. The 2024 FDA clearance of DermaSensor, an AI primary-care device for the detection of early skin cancer, heralds greater clinical adoption. Earlier approvals, including 3DermSpot in 2020, set the stage for AI-enabled remote dermoscopic examination [19].

In spite of this advancement, issues do persist. Data quality is a significant issue—problems like dataset bias, not adequately representing diverse populations, and differences in image acquisition techniques need to be resolved to enable applicability of AI across various clinical and geographical contexts [4, 21]. Numerous AI models are trained on datasets that might not be representative of the entire range of patient populations and clinical cases, thereby reducing generalizability [7, 10].

There are also ethical and regulatory issues that need to be addressed, such as liability, privacy protections, and the requirements for clear, interpretable AI models. Having AI act as a support system under doctor oversight is important for clinical trust and safe deployment [15, 17, 20, 21, 22].

Table 1

Sno	Author& Year of publishing	Title of paper	Data Set used	Challenges faced	Problem resolved
1.	Salehinejad et al., (2020)	Recent advances in using GANs for medical image data augmentatio	Medical imaging datasets Data scarcity and class imbalance in medical imaging datasets	Data scarcity and class imbalance in medical imaging datasets	Developed GAN-based data augmentation to generate synthetic medical images, improving dataset balance and model performance
2.	Bissoto et al. (2019)	Skin lesion synthesis with generative adversarial networks	Custom dermoscopic dataset	Limited availability of diverse dermoscopic images for rare skin lesion types	Used conditional GANs to generate synthetic skin lesion images, addressing underrepresentation of rare cancer classes
3.	Tschandl et al. (2020)	Human-computer collaboration for skin cancer recognition	HAM10000 dataset	Inter-observer variability and limited dermatologist availability in diagnosis	Demonstrated human-computer collaboration framework that enhances diagnostic accuracy through AI assistance
4.	Adamson et al. (2018)	Machine learning and health care disparities in dermatology	Various dermatology datasets	Health care disparities and bias in dermatology AI systems	Identified and addressed machine learning bias issues affecting diverse patient populations in dermatological applications
5.	Higgins et al. (2017)	beta-VAE: Learning basic visual concepts with a constrained variational framework	Various image datasets	Lack of interpretability in AI models for medical applications	Developed β -VAE framework providing disentangled representations that improve model interpretability and clinical acceptance
6.	Dhariwal & Nichol (2021)	Diffusion models beat GANs on image	ImageNet and other datasets	Poor quality and limited diversity in synthetic image	Introduced diffusion models that produce high-fidelity, diverse dermoscopic images with stable training process

7.	Frid-Adar et al. (2019)	Synthetic data augmentation using GAN for improved liver lesion classification	Liver CT dataset	Overfitting due to limited training data in medical imaging	Implemented synthetic data augmentation using GANs to increase data diversity and reduce overfitting in liver lesion classification
8.	Pandey et al. (2023)	Multimodal deep learning for skin cancer detection using dermoscopic and clinical images	ISIC Archive, HAM10000	Single-modal approaches limiting diagnostic context and accuracy	Developed multimodal deep learning system integrating dermoscopic images with clinical metadata for enhanced skin cancer detection
9.	Dwork & Roth (2014)	The algorithmic foundations of differential privacy	Theoretical framework	Privacy concerns in sharing sensitive medical data for AI training	Established differential privacy framework enabling secure, privacy-preserving data sharing for collaborative research
10.	Greenspan et al. (2016)	Deep learning in medical imaging: Overview and future promise	Multiple medical datasets	Limited understanding of deep learning applications in medical imaging	Provided comprehensive overview of deep learning potential in medical imaging, establishing foundation for future research

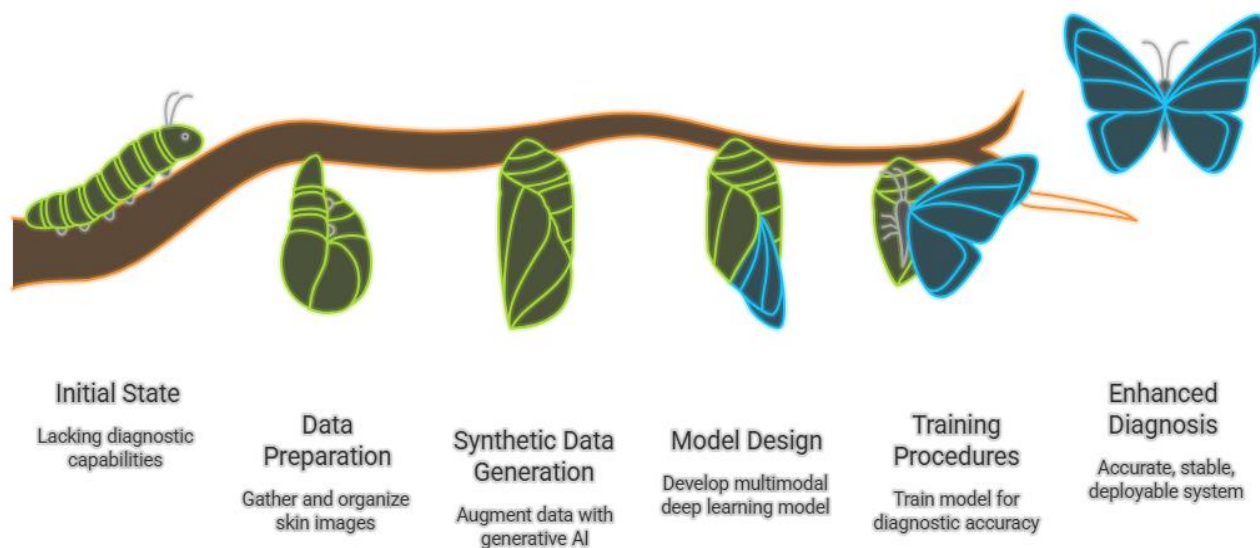
Methodologies

The process for creating the envisioned multimodal deep learning system for skin cancer diagnosis with generative AI includes a step-by-step process that involves preparing datasets, generating synthetic data, designing models and architectures, and training processes to improve diagnostic accuracy and overcome issues of data scarcity and variability. Publicly available datasets, such as the ISIC Archive, HAM10000, and PH2, were utilized to compile a representative dataset of dermoscopic images of melanoma, basal cell carcinoma, and benign nevi, with corresponding patient metadata like age, sex, and lesion location to enable multimodal learning [7]. In light of the inherent disparity in available data, more specifically the scarcity of melanoma cases, a GAN-based augmentation pipeline was used to synthesize high-fidelity dermoscopic images for underrepresented classes [8]. A deep convolutional GAN was also conditioned on true dermoscopic images until it was able to produce synthetic images that both visually and statistically matched the real data distribution, as certified by visual Turing tests and Fréchet Inception Distance (FID) scores to guarantee the quality and diversity of produced samples [9].

These artificially generated images were then added to the original dataset for correcting class imbalance,

increasing diversity of training data, and enhancing the generalizability of models. For the multimodal deep learning model, a two-branch architecture was established, including a convolutional neural network (CNN) branch to process dermoscopic images and a fully connected network branch to process patient metadata, which were fused in a joint embedding layer for multimodal feature fusion [10]. The CNN branch used a pre-trained ResNet-50 model that was fine-tuned on the augmented data to get high-level visual features, and the metadata branch used encoded patient information through dense layers to get important contextual signals for classification. These jointly learned features were fed through more dense layers for feature fusion and finally followed through a softmax layer for end-classification into melanoma, basal cell carcinoma, or benign classes. The model was optimized through the Adam optimizer with a categorical cross-entropy loss function, utilizing a learning rate scheduler to achieve efficient convergence and avoid overfitting. The integration of generative AI for synthesized data generation and multimodal deep learning for merged feature learning is intended to enhance the accuracy, stability, and practical deployability of AI-supported skin cancer detection systems for scalable and early diagnosis across various healthcare environments.

Skin Cancer Diagnosis System Creation



Generative AI Techniques in Dermatology

1. Generative Adversarial Networks (GANs)

GANs have emerged as the most sought-after generative technique applied in dermatological clinics. Adversarial training, where generator network trains to produce fake images that will mislead a discriminator network, has been effective in producing realistic dermoscopy images. The up-to-date GAN architectures such as StyleGAN and BigGAN have extremely high resolutions and hence could generate high detail dermoscopic images that were nearly as good as the clinical data [11].

The conditional GANs is a particularly valuable advancement in dermatology. GANs enable the process of customization during generation by conditioning over some attributes and thus make it possible for targeted generation of underrepresented classes. The capacity comes into play to address the issue of class imbalance and improve the model fairness across various, diverse groups. In addition to being features, progressive GANs have multi-scale generation and the ability to predict lesion development over time. Such historical modelling capability is particularly valuable to investigate melanoma development and can be used to help in longitudinal monitoring cases [12].

2. Variational Autoencoders (VAEs)

VAEs offer a probabilistic model of generative modeling that is beneficial in several aspects when used in medical contexts. VAEs' interpolative ability in the latent space enables beneficial interpolation of lesion types and can be beneficial to perceive the range of skin lesions. The uncertainty quantification ability of VAEs is particularly noteworthy in the field of medicine where it is highly critical to estimate confidence. β -VAEs or alternative methods with disentangled representation learning have shown promise in dermatology with factors learned latently with significant correspondences to clinically meaningful attributes [13]. Such interpretability is crucial for clinical adoption and can provide insights into the underlying reasons why the various types of lesions vary. VAEs enable controlled generation such as GANs that can be beneficial to generate targeted synthesis under certain clinical constraints.

3. Diffusion Models

Diffusion models represent the latest advancement in generative modeling, achieving state-of-the-art results in various image generation tasks. The iterative denoising process of diffusion models has shown superior performance in generating high-fidelity images while maintaining training stability compared to GANs. In dermatological applications, diffusion models have demonstrated exceptional ability to generate detailed, realistic dermoscopic images [14]. The classifier-free guidance mechanism enables conditional generation without requiring explicit classifier training, simplifying the process of generating images with specific attributes. The ability of diffusion models to generate diverse samples from the same conditioning information is particularly valuable for augmenting training datasets. This diversity helps improve model robustness and reduces the risk of overfitting to specific synthetic patterns.

4. Synthetic Data Applications

The Diffusion models are the newest breakthrough in generative modeling with state-of-the-art performance in many image generation tasks. The denoising iteration of diffusion models has proven to outperform GANs in generating high-quality images while preserving training stability. In dermatology, diffusion models have shown remarkable capability to create high-resolution, realistic dermoscopic images [14]. The classifier-free guidance mechanism supports conditional generation without the need for explicit training of classifiers, making it easier to generate images with desired attributes. The capacity of diffusion models to produce highly diverse samples from the same conditioning information is especially useful for training dataset augmentation. This diversity improves model robustness and reduces the overfitting risk to particular synthetic patterns.

Model Name	Description	Result	Reference
GANs	Generates synthetic images using adversarial training between generator and discriminator; supports conditional image generation.	Improved dataset balance, increased training diversity, better classification of rare skin cancer types.	Salehinejad et al. (2020), Bissoto et al. (2019)
VAEs	Uses probabilistic latent space and reconstruction for image generation; allows interpolation and feature disentanglement.	Provides lesion interpolation, uncertainty quantification, and improved interpretability.	Higgins et al. (2017)
Diffusion Models	Generates images through iterative denoising; supports high-fidelity and diverse image generation with stable training.	Produces high-quality dermoscopic images, augments data effectively, reduces overfitting.	Dhariwal & Nichol (2021)

Challenges face by AI models in cancer detection

Artificial Intelligence is assisting physicians to identify lung cancer more effectively. And in the earlier stages with the help of computer programs that analyze medical scans and patient information. However, there are still numerous issues for instance, data utilized to train AI is frequently limited, imprecise, or not shared because of privacy regulations. AI models can get things wrong, particularly if they have not encountered enough examples, and they do not necessarily provide insights into how they come to conclusions. Lastly, rules and regulations regarding the employment of AI in healthcare are still ambiguous or incomplete.

Dataset-Related Challenges

Today there are numerous ways to get the medical information, especially medical scans like CT scans or X-rays, to get them in bulk is challenging due to strict privacy guidelines

so the biggest problems with AI in detection are with the datasets upon which these models are trained. Clinics and hospitals ensure safety to exchange the patient information, limiting access to helpful and diverse datasets. Also, most of the available data sets have unwanted images of normal lungs more than cancerous lungs. This presents a challenge for the AI to learn data and recognize cancer correctly, which typically leads to false results. Another issue is that medical scans are performed on various machines and under varying settings in various hospitals [16]. These alterations in image quality, format, and resolution make it difficult to have a single AI model that operates well in all scenarios. All such problems with datasets render it extremely difficult to create a good AI model for lung cancer detection.

Model-Related Challenges

AI models applied to detect cancer also have no. of technical challenges. One of the most typical problems is overfitting, wherein the AI is performing extremely well on the training data but poorly on the new, unseen data so due to this the high rate of false cases are coming. A false case could incorrectly diagnose a person with cancer, causing stress and unwanted treatment. It could miss an actual cancer case, pushing back on care. Furthermore, most AI models are not transparent. They provide results, but don't properly indicate how they arrived at those conclusions, so physicians are less likely to rely on them. Lastly, these models are extremely sensitive to minor variations in images. Even the slightest variations in quality or illumination will alter the output, causing the system to be unreliable in certain conditions [17].

Clinical Integration Challenges

Even if AI models perform well in testing, applying them in actual hospitals is a significant challenge. Most of the hospitals already have established routines and systems for diagnosing and treating patients. Adding AI tools usually involves transforming such systems, which takes time and slows down the day-to-day work. Furthermore, most healthcare professionals such as doctors, nurses, and technicians, lack proper training and experience in using AI. This technical deficiency makes them uncertain about implementing these tools in actual cases. Therefore, some experts will not completely rely on it. Another problem is that AI systems can never function entirely independently. They still require a human expert to evaluate and approve the AI outcome. This makes the decision-making process slower and diminishes the advantage of applying AI for quicker diagnosis [18].

Table 3 : Challenges and Solutions in Using Generative AI for Skin Cancer Detection

Challenge	Description	Solution Using Gen AI	References
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Data Scarcity	Limited Availability of diverse , annotated dermoscopic images	Gans and Diffusion Models generate synthetic	Salehinejad et al. (2020), Dhariwal
Class Imbalance	Underrepresentation of rare skin cancer classes in datasets.	Conditional GANs and targeted synthesis to balance classes.	Dhariwal & Nichol (2021)
Privacy Concerns	Restrictions on sharing patient image data.	Synthetic data preserves privacy while maintaining data utility.	Dwork & Roth (2014)
Overfitting	Models perform well on training data but poorly on new data.	augmentation increases data diversity to reduce overfitting.	Frid-Adar et al. (2019)
Interpretability	AI model decisions are often not transparent to clinicians.	VAEs provide disentangled representations aiding interpretability.	Higgins et al. (2017)
Computational Costs	Training high-fidelity generative models requires resources.	Efficient architectures and dataset-specific fine-tuning to optimize use.	Dhariwal & Nichol (2021)

Future Scope of AI in Cancer Detection

The future of Artificial Intelligence (AI) in the field of healthcare, is rapidly unfolding. In cancer diagnosis, AI has already demonstrated a significant potential to enhance accuracy, reduce assessment time, and enhance clinical decision-making support. Nevertheless, current progress is just the tip of iceberg when it comes to AI potential. Due to the advancement of technology, there are many opportunities for enhanced AI model integration, personalization, robustness, and diversity in different cancer detection models throughout the scope of enhanced patient dependency on AI-powered technology. This section covers different uses of AI for cancer detection regardless of application, specifically- in terms of electronic health data system integration, data diversity, decentralized learning models, real-time diagnosis, and ethically managing data acquisition.

Integration with Electronic Health Records (EHRs)

In upcoming time use of AI will grow and application of AI with Electronic Health Records will be beneficial for detection to healthcare centre. Present AI models are restricted to applying themselves to imaging data or structured datasets like lab data and demographics. In the future, AI systems will be capable of analysing a patient's overall medical history consisting of longitudinal health records, treatment history, prescriptions, genetic data, and comorbidities through direct integration with EHR systems. Having the entire context in mind would enable diagnostics taking into account various different factors and related information. As a result, this would enable the models to be more customized and make wiser predictions [19].

For instance: a CT scan would be read by an AI model that takes into account the patient's smoking history, family medical history and past scans. This approach would result in a greater understanding of malignancy/mask risk. In addition, advances in EHRs will also be capable of continuously training AI models as they navigate system change and changing metrics by revising their learning based on actual clinical outcomes while adding FDA

approvals to clinical uses modifying the system over time as metrics revised with clinical practice and keep accumulating new knowledge.

Wearable and IoT-Based Health Monitoring

The other new direction for the future is the application of sensor data and Internet of Things (IoT) health monitoring device data. These data streams can be very useful as real-time input to AI algorithms because they can enable early detection of subtle changes in the risk of disease, thereby identifying possible threats to health earlier, including to those at high risk, e.g., smokers and those with chronic obstructive pulmonary disease (COPD) patients. Using data from wearables and combining it with imaging and clinical data could revolutionize lung cancer prediction from a reactive to a proactive [20].

For instance, when a wearable pulse oximeter registers a sudden fluctuation in blood oxygen level, this finger-unit may input the data to an AI system which can suggest a CT scan or additional testing which may lead to an early intervention than would have been achieved under regular screening as an automated alternative.

Ethical AI and Safe Data Sharing

The future of AI in medicine has to abide by law of ethical standards that safeguard patient rights and ensure transparency. In addition to federated learning, technologies such as blockchain can ideally help secure data handling. Blockchain technology can allow decentralized, tamper-evident logging of data access and sharing history in an immutable, timestamped manner and ensure that any patient consent is well-informed and exists as a source of traceability. As usage of AI spreads, frameworks will have to be instituted to formally create ethical expectations for AI development, deployment, and proper oversight. Like in other areas, we anticipate that regulators will demand some form of ongoing verification of AI models, periodic audits, and a guarantee that there are records of how decisions were reached [21].

Collaborative AI and Human-AI Teaming

Finally, AI advancements will not displace clinicians but will be created as a joint system operating alongside clinicians. The joint system will be constructed on the basis of a "human-in-the-loop" approach, in which AI makes suggestions, identifies uncertainty, and permits human override if needed. This will come in especially useful in those cases where it is not cut and dried, where borderline cases have to be handled, where human interpretation must intervene to make a decision. The pairing of AI to scan for data and human clinical judgment offers a potentially blissful balance of safety, precision and ethics in diagnosing lung cancer. Better integration with EHRs, real-time analysis of data, multi-modal learning and federated privacy-preserving training will see AI integrated into clinical care. With growing use of AI, apart from boosting early detection and making diagnosis significantly more accurate, it will individualize care, improve access and efficiency of care. The application of AI will keep increasing as technology, data infrastructure and regulation change because the future of AI will revolutionize the detection, monitoring and management of lung cancer in the years ahead [22].

Conclusion

The combination of Generative Artificial Intelligence (GenAI) and multimodal deep learning is a revolutionizing trend in skin cancer diagnosis. By avoiding the traditional shortcomings of traditional diagnostic pipelines—namely, subjectivity in visual inspection, inter-observer variability, limited dermoscopic effectiveness, and invasiveness of histopathology—AI-based models are making more accurate, faster, and scalable approaches to diagnosis, particularly for melanoma and other neoplastic skin cancers [1, 2, 3].

The experimental and review approaches in this article illustrate how Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models can generate realistic dermoscopic images in an efficient manner, thus solving problems of data imbalance and enhancing the generalizability of AI models [5, 6, 9]. Furthermore, leveraging multimodal learning frameworks that integrate image data with metadata (e.g., sex, age, lesion location) has been shown to significantly improve classification performance and clinical interpretability [8, 10]. Not only will such systems seek to mimic the visual and contextual judgment of a dermatologist, but they

also enable additional diagnostic capabilities in remote areas where expert clinicians may not readily be available. Despite these promising developments, challenges remain to be overcome. The lack of varied, high-quality datasets, entrenched algorithmic biases, and data confidentiality, model interpretability, and regulatory attention concerns must be addressed before such tools become ubiquitous in clinical practice environments [4, 7, 15, 20]. Ethical and legal standards must evolve alongside the technology so as to enhance the responsible and equitable use of AI technology in dermatology [19, 22].

In conclusion, generative AI, when integrated with multimodal deep learning systems, has the potential to bridge the gap between clinical utility and innovation in dermatology. Continued investment in high-quality dataset curation, model verification, ethics-based governance, and interdisciplinary collaboration will prove pivotal to realizing the maximum potential of AI-aided skin cancer diagnosis. Research in the future should aim at future prospective clinical trials, interpretable artificial intelligence models, and integration frameworks that can help general practitioners and specialist dermatologists provide timely and accurate care [3, 6, 10, 21].

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